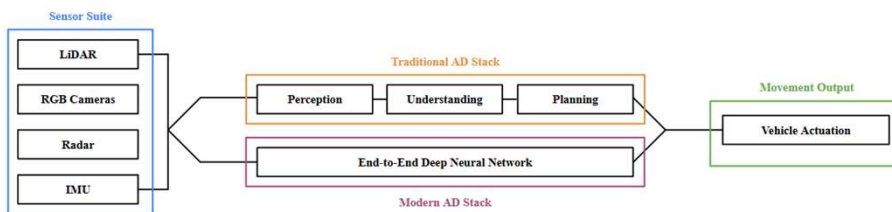


Simulation Frameworks for Automated Driving

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1. Introduction

Autonomous driving (AD) is a rapidly developing field, of which a large amount of research is fixated on. For a vehicle to navigate an environment autonomously, it must be aware of its surroundings. This is typically accomplished through a multi-sensor structure, allowing for input data to be collected and used to control vehicle movement downstream. However, the data-processing pathway that results in actuation differs depending on the approach. While traditional methods compartmentalize the task into various modules, the drastic increase in artificial intelligence systems has sought to instead implement end-to-end machine learning architectures. Because of this, a large amount of data is needed for the training and validation of models, yet the data can be obtained through many ways, each introducing unique challenges.

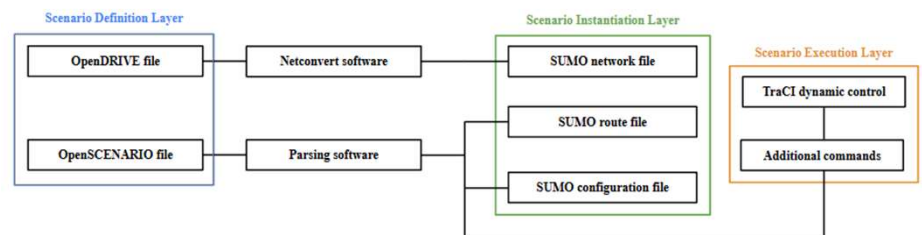


2. Project Description

This project aims to develop end-to-end automated driving systems. Within this broad scope, the research aims specifically to collect data within simulated environments, allowing for the testing and validation of models. Simulation achieves this in a much safer and cost-effective manner than that of real-world methods, but it also has its downsides. There are a variety of simulators available to developers, all tailored to providing certain characteristics. Within this project, the applications Simulation of Urban Mobility (SUMO), Car Learning to Act (CARLA), and Environment Simulator Minimalistic (esmini) have been implemented. The multi-faceted approach enables the emulation of real-world traffic flow while also maintaining accurate movement dynamics for vehicles of interest, allowing for the collection of high-fidelity data of a driving environment.

3. Conducted Research

The focus of this project is to build upon pre-existing support for co-simulation approaches. These frameworks use OpenSCENARIO to define driving instances in CARLA and utilize SUMO for background traffic behavior. However, a gap in this approach is the compatibility between the architectures of OpenSCENARIO and SUMO. Unifying these structures is of interest for data collection efforts, and has been explored throughout the duration of this project.



5. Conclusion

The work produced from this project has initiated the compatibility between OpenSCENARIO and SUMO structures, with direct relevance to data collection methods. This serves as a foundation for continued development towards the coordinated approach of simulating realistic driving environments. High-fidelity simulation data allows for the training of end-to-end models that can be tested and developed safely within virtual driving environments. Once virtual development is complete, the models created in simulation may be implemented into reality through physical automobile fleets. For more details of this project, as well as the source code, please scan the QR code below.



4. Results

A software pipeline has been developed to recreate scenarios defined in OpenSCENARIO within the SUMO environment. This has been done for nine categorized instances across a range of road networks, as well as numerous route-based sequences employing high-precision road data via OpenStreetMap. To ensure the accurate reproduction of OpenSCENARIO events in SUMO, all scenes were run on their respective software's graphical user interface, allowing for visual verification of identical behavior.

